ELECTRONICS AND COMMUNICATION ENGINEERING

COMMUNICATION SYSTEM



Baseband Transmission

Lecture No.:3

By- SAKET VERMA SIR



1. Types of Sampling



1. Practice Questions
2. NAT Questions
3. Bandpass sampling

FOLLOW US ON

TELEGRAM

https://t.me/Elexbysvsir

SAKET VERMA SIR



$\eta=1 \quad X(f-\eta f_s) \quad f_s \pm f_m = 10 \pm 4 = 6 \& 14$

#Q. Consider a sampled signal $y(t) = 5 \times 10^{-6}\, x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$ where $x(t) = 10 \cos(8\pi \times 10^3\, t)$ and $T_s = 100\ \mu sec$. When $y(t)$ is passed through an ideal low pass filter with a cut off frequency of 5 KHz, the output of the filter is

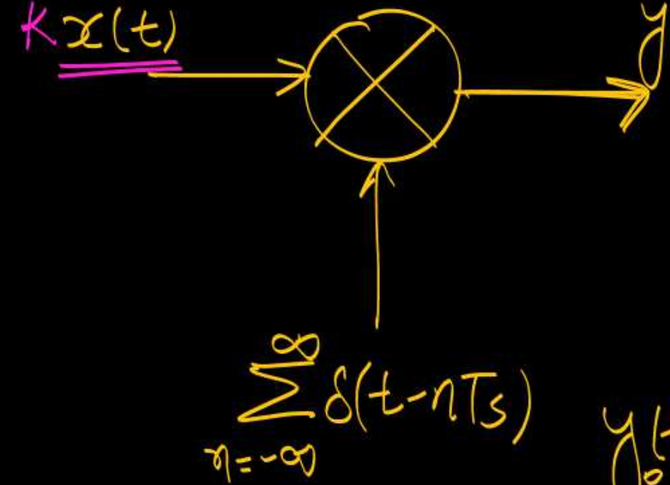
f_m: 4 KHz

(A) $5 \times 10^{-6} \cos(8\pi \times 10^3 t)$

(B) $5 \times 10^{-5} \cos(8\pi \times 10^3 t)$

(C) $5 \times 10^{-1} \cos(8\pi \times 10^3 t)$

(D) $10 \cos(8\pi \times 10^3 t)$



$y(t) \longleftrightarrow \frac{K}{T_s} \sum X(f-\eta f_s)$

$\longleftrightarrow \frac{1}{100\times10^{-6}} X(f-\eta f_s) \times K$

$y_o(t) \longleftrightarrow 10^4\, X(f) K \quad for\ n=0$

$y_o(t) = K10^4\, x(t) = K10^4 \times 10 \times \cos(8\pi \times 10^3 t)$

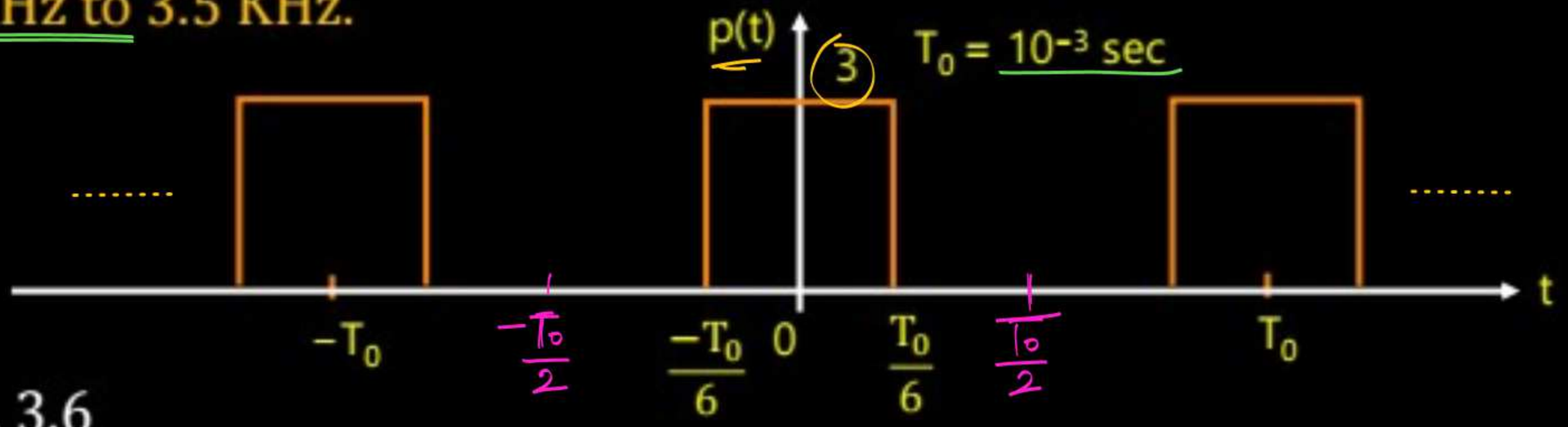
$= 10^5 K \cos(8\pi \times 10^3 t)$

$y_o(t) = 10^5 \times 5 \times 10^{-6} \cos(8\pi \times 10^3 t)$

$= 5 \times 10^{-1} \cos(\,.\,)$

#Q. Let $x(t) = 2\cos(800\pi t) + \cos(1400\pi t)$, x(t) is sampled with the rectangular pulse train shown in figure below. The only spectral components (in KHz) present in the sampled signal in the frequency range 2.5 KHz to 3.5 KHz.

multitone



$f_s = \frac{1}{T_0} = 1$ KHz

(A) 2.7, 3.4

(B) 3.3, 3.6 ✗

(C) 2.6, 2.7, 3.3, 3.6

(D) 2.7, 3.3 ✓

$f_1 = 0.4\ KHz, \quad f_2 = 0.7\ KHz$

$f_s \pm f_1 = 1 \pm 0.4 = 1.4 \text{ & } 0.6$

$2f_s \pm f_1 = 2 \pm 0.4 = 2.4 \text{ & } 1.6$

n=3 $3f_s \pm f_1 = 3 \pm 0.4 = 3.4 \text{ & } 2.6$

$4f_s \pm f_1 = 4 \pm 0.4 = 4.4 \text{ & } 3.6$

$f_s \pm f_2 = 1 \pm 0.7 = 0.3 \text{ & } 1.7$

$2f_s \pm f_2 = 2 \pm 0.7 = 1.3 \text{ & } 2.7$

$3f_s \pm f_2 = 3 \pm 0.7 = 3.7 \text{ & } 2.3$ ✗

$4f_s \pm f_2 = 4 \pm 0.7 = 3.3 \text{ & } 4.7$

2 to 4

$p(t)$ is a periodic signal with $T_0 = 10^{-3}$. We can write TFS,

$$\omega_0 = \frac{2\pi}{T_0}$$

$$a_0 = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} p(t)\,dt = \frac{1}{T_0}\int_{-T_0/6}^{T_0/6} 3\,dt = \frac{3}{T_0}\left(\frac{T_0}{3}\right) = 1$$

$$a_n = \frac{2}{T_0}\int_{-T_0/2}^{T_0/2} p(t)\cos n\omega_0 t\,dt = \frac{2}{T_0}\int_{-T_0/6}^{T_0/6} 3\cos n\omega_0 t = \frac{12}{T_0}\int_{0}^{T_0/6} \cos n\omega_0 t\,dt$$

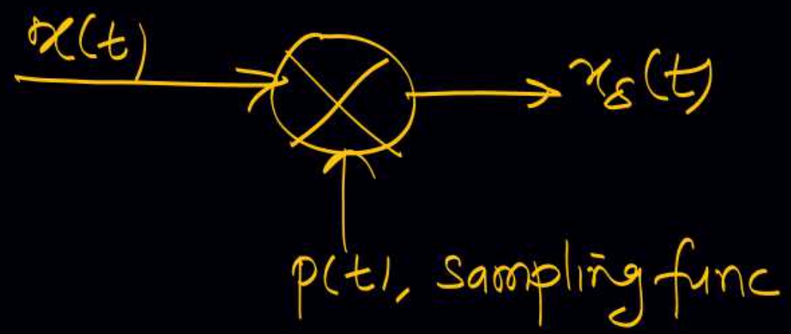
$$= \frac{12}{n\omega_0 T_0}\sin\left(\frac{n\omega_0 T_0}{6}\right) = \frac{12}{n2\pi}\sin\left(\frac{2n\pi}{6}\right) = \frac{6}{n}\sin\left(\frac{n\pi}{3}\right).$$

$p(t)$ is even signal,

$$b_n = 0.$$

$$p(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t)$$

$$= 1 + \sum_{n=1}^{\infty} \frac{6}{n\pi} \sin\left(\frac{n\pi}{3}\right) \cos(n\omega_0 t)$$



$$x_s(t) = x(t)\,p(t) = \left[2\cos(800\pi t) + \cos(1400\pi t)\right]\left[1 + \sum_{n=1}^{\infty} \frac{6}{n\pi} \sin\left(\frac{n\pi}{3}\right) \cos(n\omega_0 t)\right]$$

$\sin n\pi$, $n=3$, $= 0$

$n=0$, $f_1 = 0.4$ KHz & $f_2 = 0.7$ KHz

$n=1$, $f_0 \pm f_1 = 1 \pm 0.4 = 1.4$ & 0.6 KHz $f_0 \pm f_2 = 1 \pm 0.7 = 1.7$ & 0.3 KHz.

$n=2$ $2f_0 \pm f_1 = 2 \pm 0.4 = 2.4$ & 1.6 KHz $2f_0 \pm f_2 = 2 \pm 0.7 = 2.7$ & 1.3 KHz

$n=3$ $3f_0 \pm f_1$ O/P = 0

$n=4$ $4f_0 \pm f_1 = 4 \pm 0.4 = 3.6$ & 4.4 $4f_0 \pm f_2 = 4 \pm 0.7 = 3.3$ & 4.7 KHz.

$f_m = 0.5\ KHz$

$$g(t) = -g\left(t \pm \frac{T_s}{2}\right)$$

HWS

#Q. A signal with bandwidth 500 Hz is first multiplied by a signal g(t) where

$$g(t) = \sum_{k=-\infty}^{\infty} (-1)^k \delta(t - 0.5 \times 10^{-4}k)$$

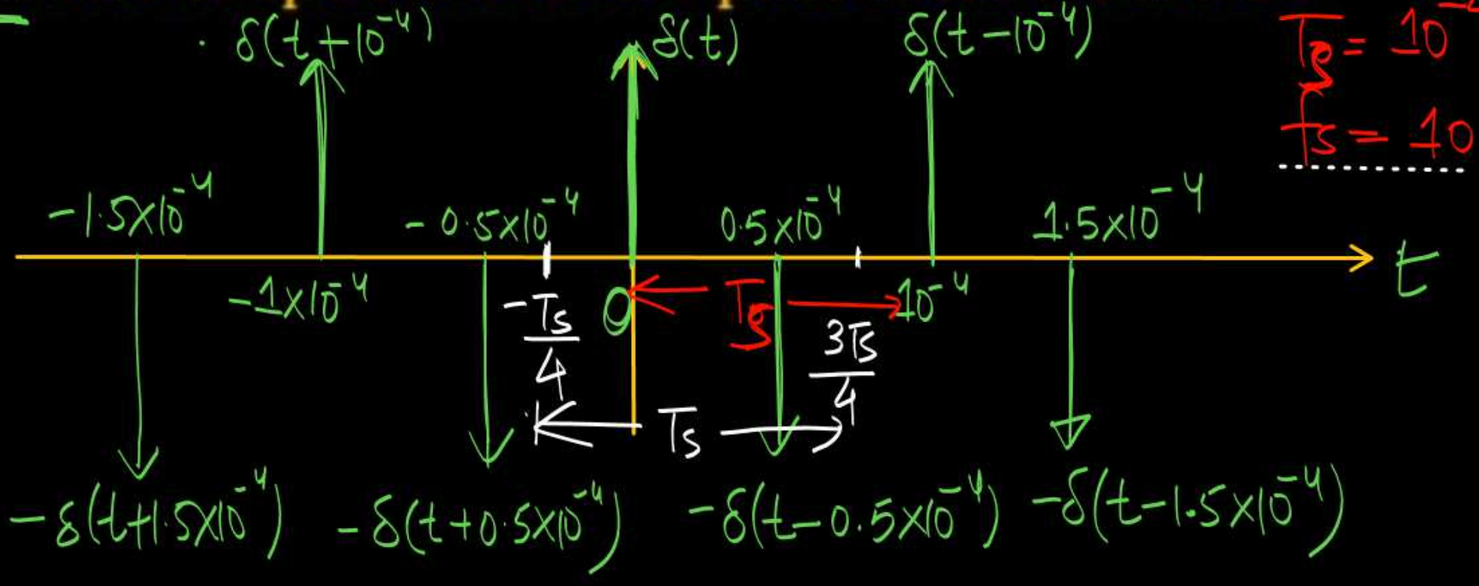
The resulting signal is then passed through an ideal low pass filter with bandwidth 1 KHz. The output of the low pass filter would be

(A) $\delta(t)$

(B) $m(t)$

(C) 0

(D) $m(t).\delta(t)$



$T_s = 10^{-4}$ sec.

$f_s = 10$ KHz.

$$a_0 = \frac{1}{T_s}\int_{-T_s/2}^{T_s/2} g(t)\,dt = \frac{\text{Area over one cycle}}{T_s} = \frac{1-1}{T_s} = 0$$

Due to even HWS. odd harmonics cosine will be present in $g(t)$.

$f_s,\ 3f_s,\ 5f_s \ldots\ldots$

$$g(t)\,m(t) = m(t)\left[K_1\cos(2\pi f_s t) + K_2\cos(2\pi 3f_s t) + K_3\cos(2\pi 5f_s t) + \ldots\right]$$

$f_m = 0.5\,KHz$

Ans.

o/p of LPF = 0

After sampling freq. components present,

$f_s \pm f_m,\quad 3f_s \pm f_m,\quad 5f_s \pm f_m \ldots\ldots$

10 ± 0.5 — 9.5 & 10.5

30 ± 0.5 — 29.5 & 30.5

50 ± 0.5 ....

$f_m = 5\,Hz, \quad f_s = 15\,Hz.$

$A\,T\,\text{sinc}(tT) \longleftrightarrow A\,\text{rect}\left(\frac{f}{T}\right)$

#Q. The signal $\cos\left(10\pi t + \frac{\pi}{4}\right)$ is ideally sampled at a sampling frequency of 15 Hz. The sampled signal is passed through a filter with impulse response $\left[\frac{\sin \pi t}{\pi t}\right] \cos\left(40\pi t - \frac{\pi}{2}\right)$.

BPF

$\text{sinc}(t) \cdot \cos(40\pi t - \pi/2) = \underbrace{\text{sinc}(t)}_{x(t)} \sin(40\pi t)$

(A) $\frac{15}{2}\cos\left(40\pi t - \frac{\pi}{4}\right)$

(B) $\frac{15}{2}\cos\left(10\pi t + \frac{\pi}{4}\right)\left[\frac{\sin \pi t}{\pi t}\right]$

(C) $\frac{15}{2}\cos\left(10\pi t + \frac{\pi}{4}\right)$

(D) $\frac{15}{2}\cos\left(10\pi t - \frac{\pi}{4}\right)\left[\frac{\sin \pi t}{\pi t}\right]$

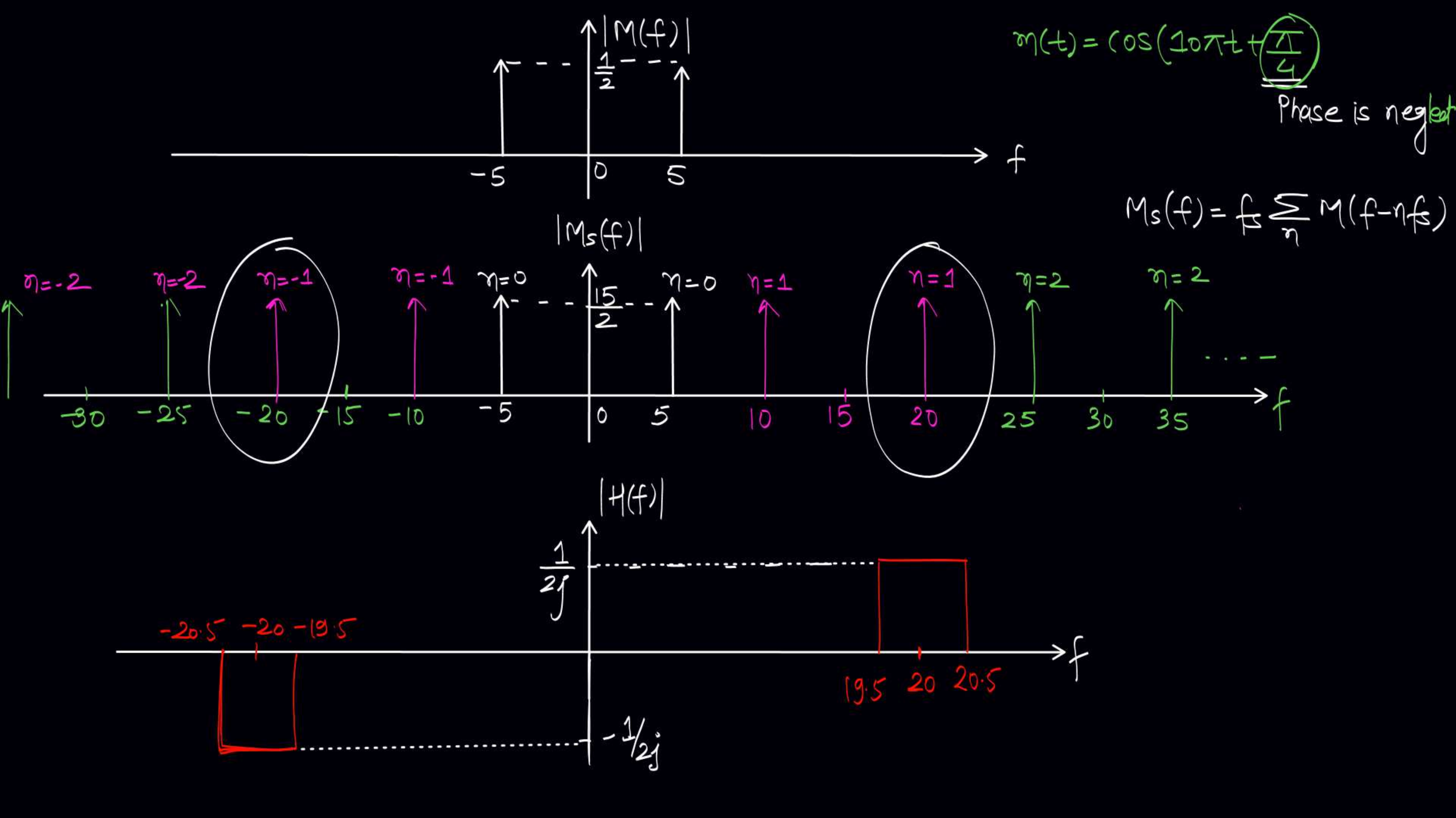
$x(t) \longleftrightarrow X(f)$

$x(t)\sin(2\pi f_c t) \longleftrightarrow \frac{1}{2j}\left[X(f-f_c) - X(f+f_c)\right]$

$\text{sinc}(t) \longleftrightarrow \text{rect}(f)$



$\text{sinc}(t)\sin 2\pi f_c t \longleftrightarrow \frac{1}{2j}\left[\text{rect}(f-20) - \text{rect}(f+20)\right]$


|M(f)|
1/2
-5
0
5
f
m(t) = cos(10πt + π/4)
Phase is neglected
Ms(f) = fs Σ_n M(f - nfs)
|Ms(f)|
n=-2
n=-2
n=-1
n=-1
n=0
n=0
n=1
n=1
n=2
n=2
15/2
-30
-25
-20
-15
-10
-5
0
5
10
15
20
25
30
35
f
|H(f)|
1/2j
-20.5
-20
-19.5
19.5
20
20.5
f
-1/2j

Output of filter,

$$Y(f) = \frac{1}{2j}\left[\frac{15}{2}\delta(f-20) - \frac{15}{2}\delta(f+20)\right]$$

$$= \frac{15}{2}\left[\frac{1}{2j}\{\delta(f-20) - \delta(f+20)\}\right]$$

Taking IFT,

$$y(t) = \frac{15}{2}\sin(2\pi \times 20t) = \frac{15}{2}\cos\left(40\pi t - \frac{\pi}{2}\right)$$

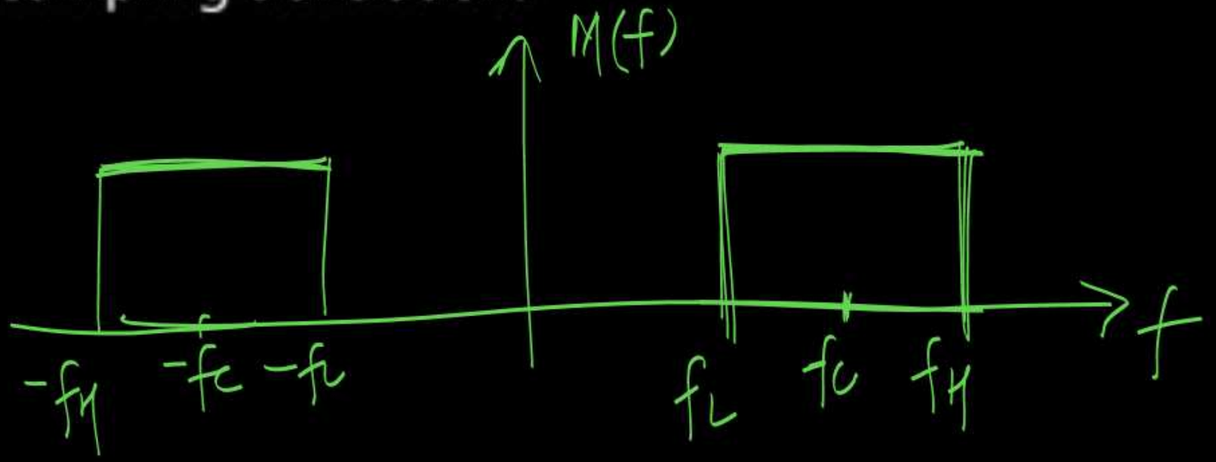
Considering the neglected phase,

$$y(t) = \frac{15}{2}\cos\left(40\pi t - \frac{\pi}{2} + \frac{\pi}{4}\right) = \frac{15}{2}\cos\left(40\pi t - \frac{\pi}{4}\right)$$



Bandpass Sampling

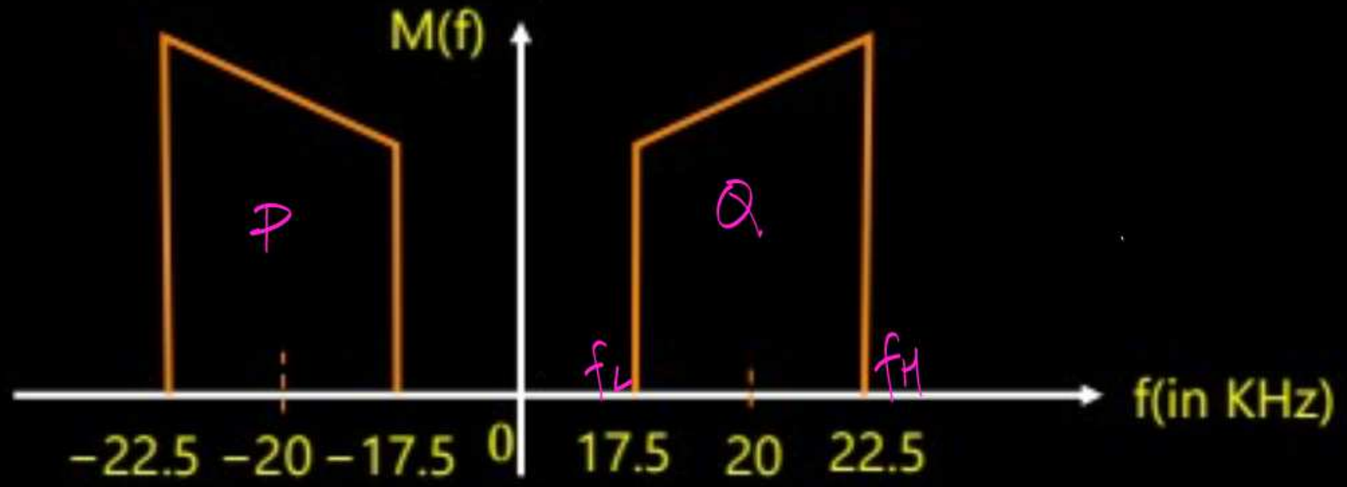
- It is also known as Harmonic Sampling, Sub – Nyquist sampling and Under sampling. ($f_s \leq 2f_H$)
- In bandpass sampling, we are more concerned with signals bandwidth than its highest frequency component.
- Band pass sampling performs digitalization and frequency translation in a single process often called as sampling translation.



Bandpass Sampling

$\frac{B}{2} = 2.5$ KHz

Consider a band pass signal m(t) of bandwidth B = 5 KHz having its spectrum as shown below:-



f_c = 20 KHz

f_H = 22.5 KHz

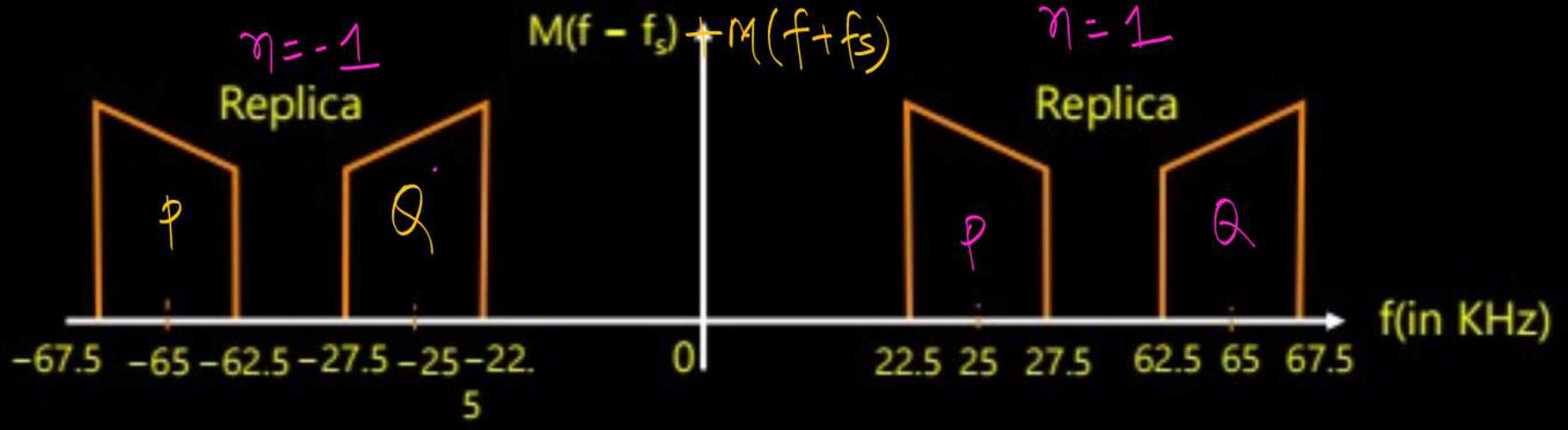
$f_s = 2f_H$ = 45 KHz

Bandpass Sampling

$\sum M(f - nf_s)$

$M(f - nf_s)$ $\quad n = 0;\ M(f) \quad n = 1;\ M(f - f_s)$

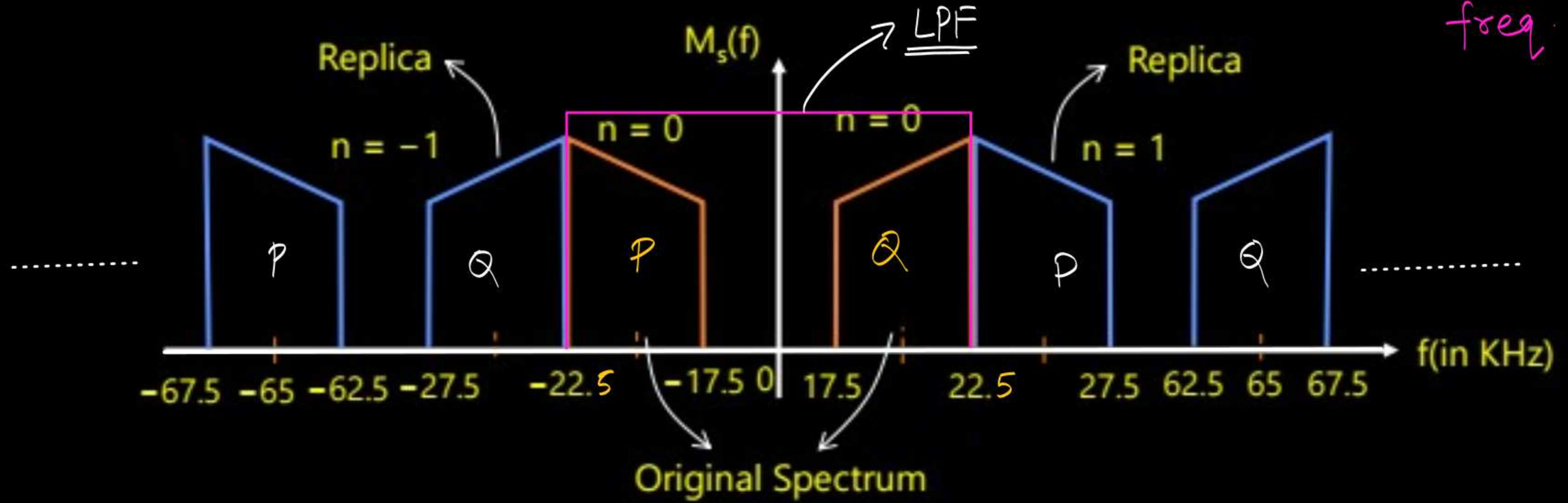




Bandpass Sampling

$M_s(f) = M(f) + M(f - f_s) + M(f + f_s) + \ldots\ldots$

$f_s = 2f_H$ Nyquist Sampling freq.





Bandpass Sampling

$m=0, \quad 45 \le f_s < \infty$

The sampling frequency f_s is given by:-

$$\frac{2f_c + B}{m+1} \le f_s \le \frac{2f_c - B}{m}$$

So that $f_s \ge 2B$

where k = m + 1

$k \rightarrow \frac{f_H}{B}$ (integer part only)

m = Number of Replica

→ 8.9 K=8

→ 2.5 K=2

→ 3.01 K=3

NOTE: If $f_c \gg B$ i.e., Narrowband Signal or Nyquist Sampling

$$f_s = 2f_H$$

given:- $f_c = 20\,KHz, \quad B = 5\,KHz$

To Calculate min. sampling freq.

Step1:- Bandwidth, $B = f_H - f_L$

Step2:- $K \rightarrow \frac{f_H}{B}$ (integer part)

Step3:- $(f_s)_{min} = \frac{2f_H}{K}$

Sol:- $B = 5\,KHz$

$K \rightarrow \frac{22.5}{5} = 4.5$

$K = 4$

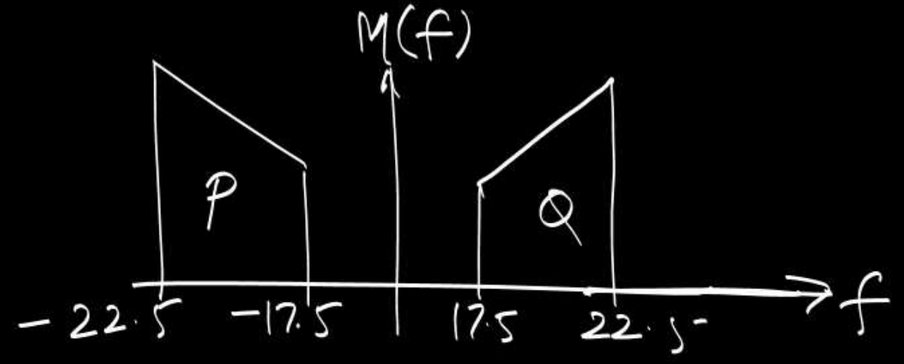
$(f_s)_{min} = \frac{2f_H}{K} = \frac{2 \times 22.5}{4} = 11.25\,KHz$

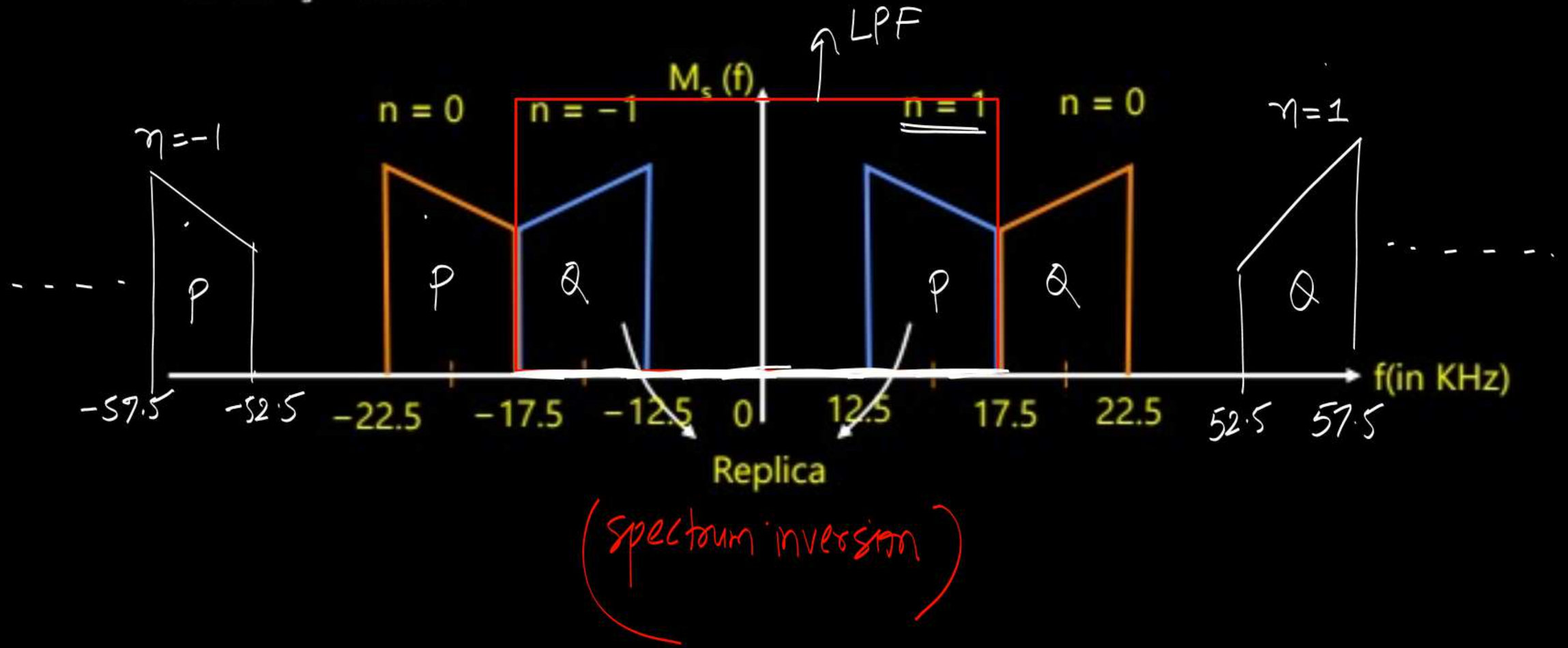
Bandpass Sampling

no. of replica between original spectrum.

CASE 1: m = 1 $22.5\ \text{KHz} \le f_s \le 35\ \text{KHz}$

When $f_s = 35$ kHz



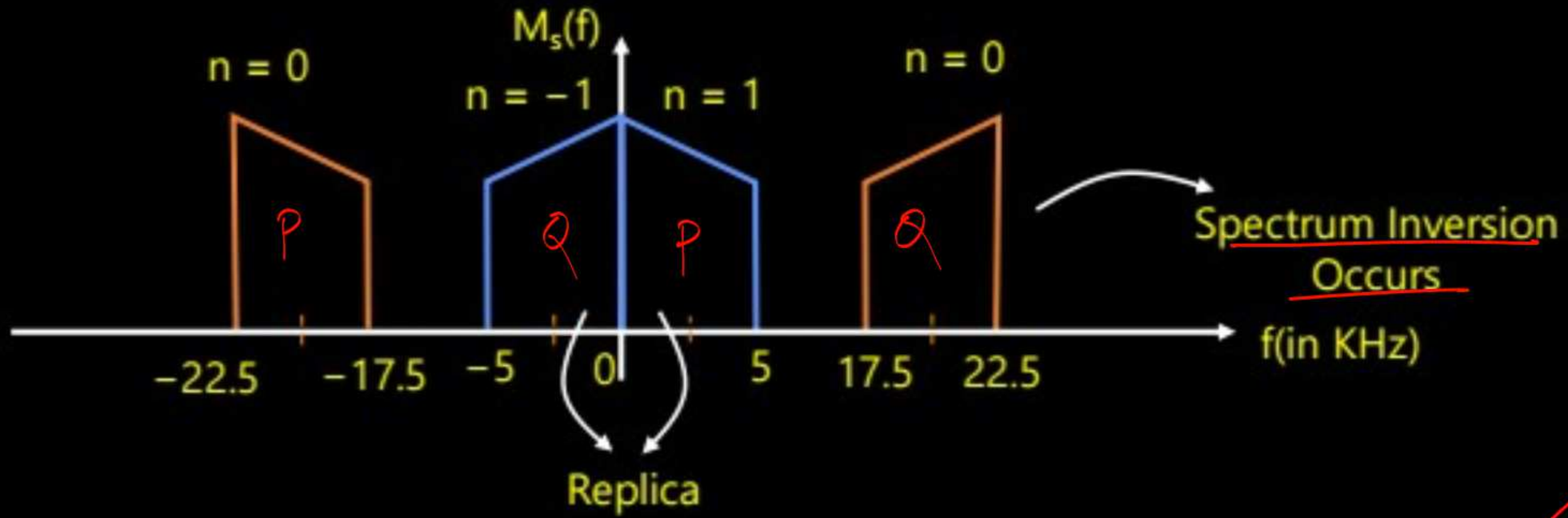


(spectrum inversion)

Bandpass Sampling

CASE 1: m = 1 $22.5\ \text{KHz} \leq f_s \leq 35\ \text{KHz}$

When $f_s = 22.5\ \text{KHz}$

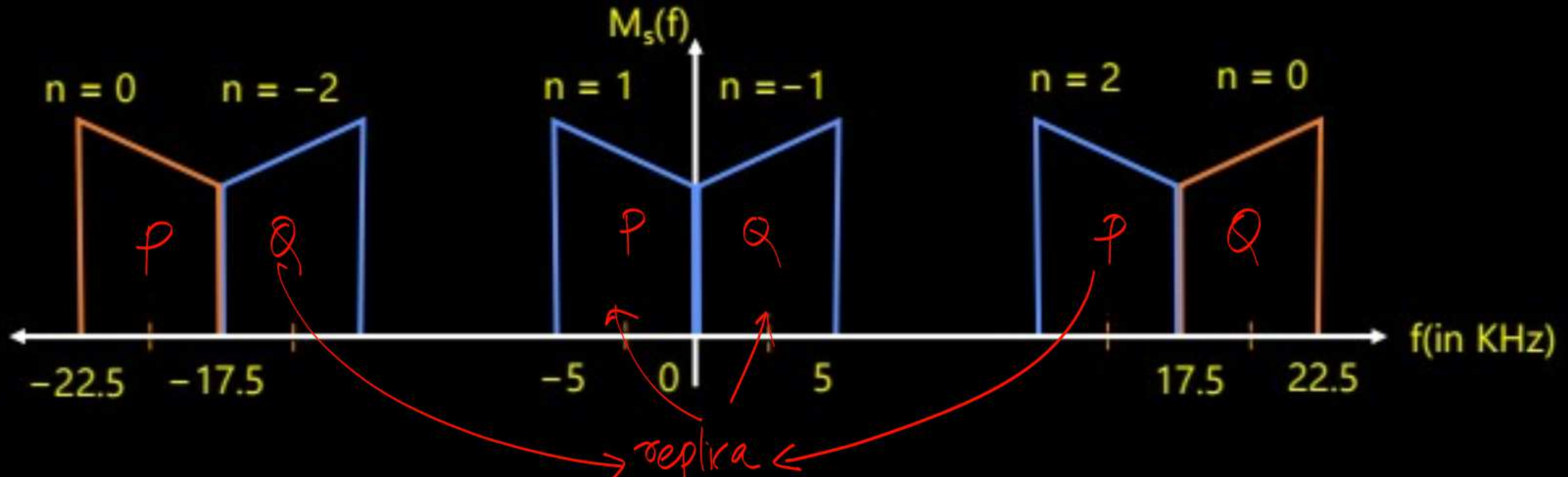


Message can be obtained by using LPF, so optimum sampling frequency = 22.5 kHz

Bandpass Sampling

CASE 2: m = 2 (even) $15\ \text{KHz} \le f_s \le 17.5\ \text{kHz}$

When $f_s = 17.5\ \text{kHz}$

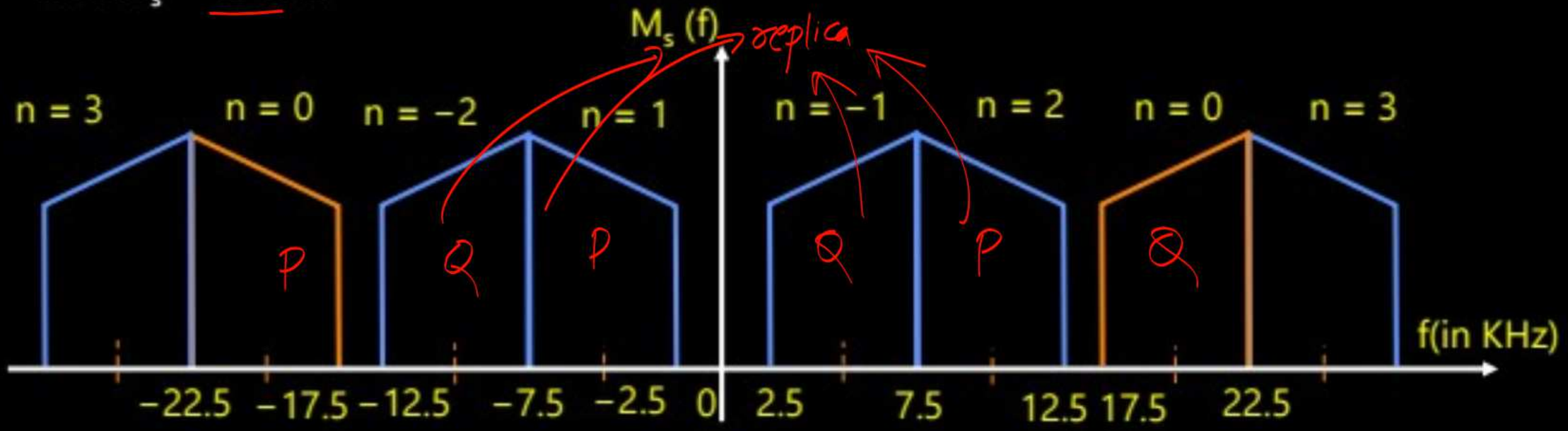


∴ Message signal can be obtained by using LPF and No spectrum Inversion.

Bandpass Sampling

CASE 2: m = 2 $15\ KHz \leq f_s \leq 17.5\ kHz$

When f_s = 15 kHz



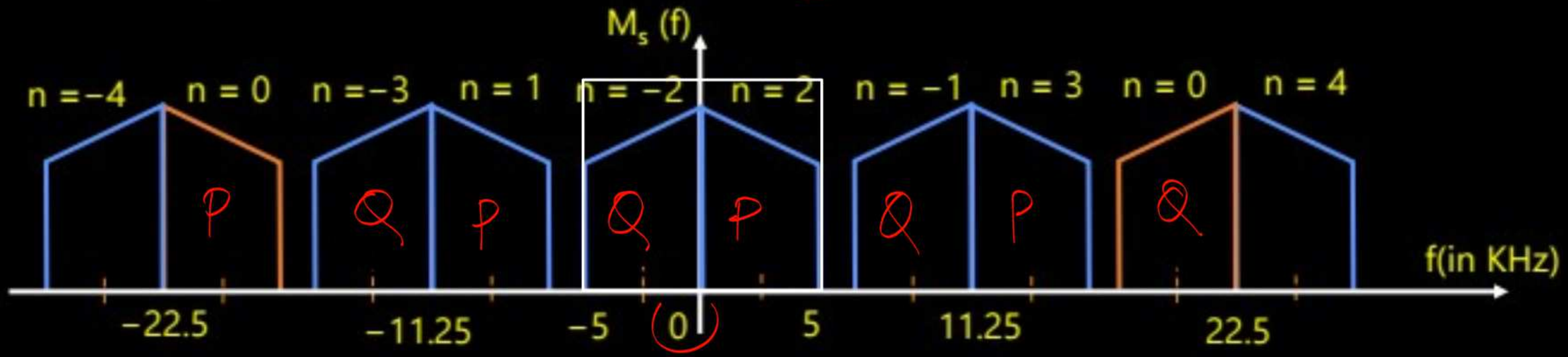
$\therefore f_s$ = 17.5 KHz is more efficient, so Band Pass Filter is not Needed.

This process is continued till $f_s \geq 2B$

Bandpass Sampling

CASE 3: $m = 3$ $11.25\ \text{KHz} \leq f_s \leq 11.67\ \text{kHz}$

When $f_s = 11.25$ kHz



∴ Replica is repeated after every f_s interval Spectrum Inversion occurring.

$f_s = 11.25$ kHz is the optimum sampling frequency.



Bandpass Sampling

CASE 4: $m = 4$ $\quad 9\ KHz \leq f_s \leq 8.75\ kHz$

$f_s \geq 2B$

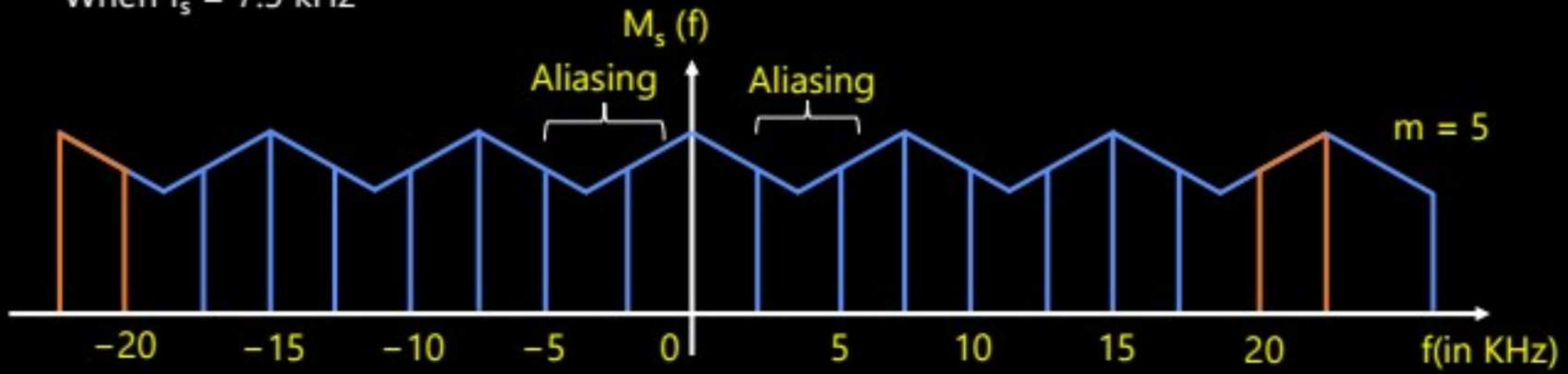
But in this case $f_s < 10$

Both equations are not satisfied, So overlapping Will occur.

Bandpass Sampling

CASE 5: $m = 5$ $\quad 7.5\ KHz \leq f_s \leq 7\ kHz$

When $f_s = 7.5$ kHz



$f_s \geq 2B$

But in this case $f_s < 10$

Both equations are not satisfied, So overlapping Will occur.



Bandpass Sampling

Conclusions :

To avoid Spectrum inversion m will be even integer.

m	$\frac{2f_c + B}{m+1}$	$\frac{2f_c - B}{m}$	Optimum f_s
1.	22.5 KHz	35 KHz	22.5 KHz
2.	15 KHz	17.5 KHz	17.5 KHz
3.	11.25 KHz	11.67 KHz	11.25 KHz
4.	9 KHz	8.75 KHz	–
5.	7.5 KHz	7 KHz	–

valid min. sampling freq.

THANK you